```
#include <stdio.h>
#include<stdlib.h>
#define TABLE_SIZE 10
int h[TABLE_SIZE]={NULL};
void insert()
{
 int key,index,i,flag=0,hkey;
 printf("\nenter a value to insert into hash table\n”);
 scanf("%d",&key);
 hkey=key%TABLE_SIZE;
 for(i=0;i<TABLE_SIZE;i++) {
     index=(hkey+i)%TABLE_SIZE;
     if(h[index] == NULL) {
        h[index]=key;
        break;
      }
    }
    if(i == TABLE_SIZE)
       printf("\nelement cannot be inserted\n");
}
void search()
{
 int key,index,i,flag=0,hkey;
 printf("\nenter search element\n");
 scanf("%d",&key);
 hkey=key%TABLE_SIZE;
 for(i=0;i<TABLE_SIZE; i++)
 {
     index=(hkey+i)%TABLE_SIZE;
     if(h[index]==key)
    {
       printf("value is found at index %d",index);
       break;
     }
    }
    if(i == TABLE_SIZE)
       printf("\n value is not found\n");
}
void display() {
  int i;
  printf("\nelements in the hash table are \n");
  for(i=0;i< TABLE_SIZE; i++)
  printf("\nat index %d \t value = %d",i,h[i]);
}
```

```
int main() {
    int opt,i;
    while(1) {
        printf("\nPress 1. Insert\t 2. Display \t3.Search \t4.Exit \n");
        scanf("%d",&opt);
        switch(opt) {
          case 1:
              insert();
              break;
          case 2:
              display();
              break;
          case 3:
              search();
              break;
          case 4:exit(0);
        }
     }
}
```